MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RADIAL BEAM SWITCHING

TYPE 6710

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Internal permanent magnets, high vacuum, 10 outputs

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS: See note 8

Parameter:	Ef	Eta	Esp	Esg	Ecc	Rta	Rsp
Unit:	V	Vdc	Vdc	Vdc	Vdc	Ohms	Ohms
Maximum:	6.9	300	100	200	200	---	---
Minimum:	5.7	---	---	---	---	---	---
TEST CONDITIONS:	6.3	55	55	55	40	3.3 K ±1% (Note 10)	150 K ±1%

ABSOLUTE RATINGS: See note 8

Parameter:	-Esig	Ehk	F	TA	P/ta	Ik	Alt
Unit:	V	V	MHz	°C	W	mA	ft
Maximum:	---	+100	Note 7	+85	1	10	60,000
Minimum:	---	-100	Static	-65	---	---	---
TEST CONDITIONS:	-55 (Notes 1, 6)	---	---	---	---	---	---

GENERAL:

(B) Qualification - Not required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1						
---	Counting voltage range (1)	3, 4, 5, 9, 11	4 Hz	Eta, Esp, Esg	49	61	Vdc
---	Counting voltage range (2)	3, 4, 5, 9, 11	10 kHz	Eta, Esp, Esg	49	61	Vdc
4250	Electrode current (target)	---	Individual targets	Ita	1.9	3.5	mA
	Quality conformance inspection, part 2						
---	Firing voltage	12, 18	Eta = Esp = Esc = 100 Vdc, Esig = 0	---	---	---	---
---	Counting voltage range (3)	3, 5, 9, 11, 13	1 kHz; Esig = -60 V	Eta, Esp, Esg	59	61	V
1001	Barometric pressure, reduced	---	55 +5 deg, 300 Vdc between targets and all other elements	---	---	---	---
1051	Heater current	---		If	135	165	mA
1030	Heater-cathode leakage current	---	Ehk = +100 V	Ihk	---	20	μA
---	Shock	14	30 G, 11 ±1 ms duration	---	---	---	---
1031	High-frequency vibration	1, 4, 15, 17	10 kHz	---	---	---	---
---	Shock and high-frequency vibration end points	16		---	---	---	---
2102	Envelope stress	---		---	---	---	---
1101	Permanence of marking	---		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test (1) (Dynamic)	---	Group C, 120 counts/sec sine wave excitation, t = 1,000 hours (min)	---	---	---	---
---	Life test (2) (Static)	---	Group C, beam formed and locked on any one target; t = 1,000 hours (min)	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3 -Continued						
---	Life-test (1) and (2) end points:						
---	Target current (individual)	---		Ita	1.7	3.8	mA
---	Counting voltage range (1)	---		Eta, Esp, Esg	52	61	Vdc
---	Counting voltage range (2)	---		Eta, Esp, Esg	52	61	Vdc
(8) ---	Electrical noise	2, 9, 19	Individual targets	---	---	20	mVac

NOTES:

1. In order to avoid possible operational changes in the characteristics of this tube, do not mount on a ferrous chassis, or operate tube less than 2.5 inches center to center.
2. Noise is measured across Rta with an ac Electronic Voltmeter (Ballantine Model No. 300, or equivalent).
3. Tube shall count 10 positions at specified frequency and throughout counting voltage range.
4. Grid swing from +40 to -15 V with respect to cathode. Only one set of grids shall be negative with respect to the grid bias (Ecc) at any time. See figure 3 for appropriate wave shape.
5. Individual spade stem leads capacity should not exceed 10 pF with tube in socket.
6. In order to avoid permanent changes in electrical characteristics, do not strike tube with other magnetic materials, or subject tube to strong magnetic fields.
7. Tube is tested up to 1 MHz but may operate to 2.0 MHz under optimum circuit conditions.
8. Following are the definitions of symbols used in this tube specification sheet.

   Rta - Resistance in series with target.
   Rsp - Resistance in series with spade.
   Esp - Spade voltage.
   Esg - Shield grid voltage.

9. Test per circuit in figure 2.
10. This is the total resistance in series with each of the 10 targets and 10 spades.
11. Counting voltage ranges should be read from center point down for minimum limit, and center point up for maximum limit.
12. Opening the switch (SW1) in figure 2 for minimum of 1 millisecond should clear the beam, closing switch (SW1) should not reform the beam.
13. Grid swing from +40 to -20 V with respect to cathode. Only one set of grids shall be negative with respect to the grid bias (Ecc) at any time. See figure 4 for appropriate wave shape.
14. Shock test shall be made on any shock machine capable of producting a half-wave shock waveform of the specified duration and amplitude. No voltages shall be applied during this test. Each tube shall be subjected to a total of 20 shocks, that is, 5 shocks in each of positions X, Y, Z plus, and Z minus in any sequence. This is a destructive test. Tubes used in the destructive test shall not be shipped on the contract or order.
15. Tap short test does not apply.
16. The tube shall count 10 at the specified test conditions at 10 kHz.
17. The tube shall count 10 at specified frequency.
18. The AQL for "Firing voltage" shall be 2.5, inspection level II.

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NOTES: -Continued

19. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 99
DSA - ES

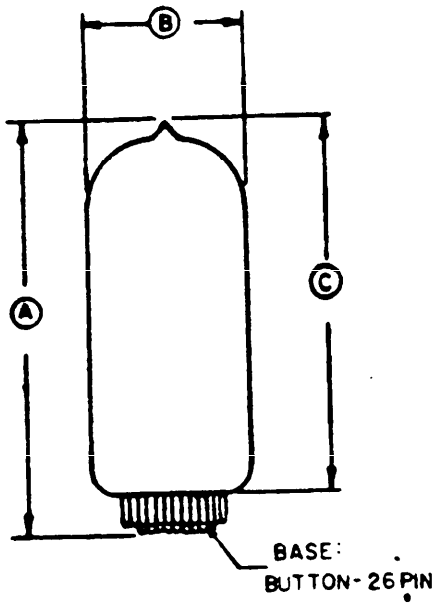
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Agent:
DSA - ES

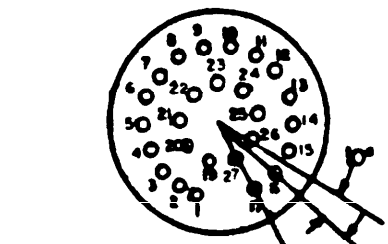
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		3.125(79.38)
B		1.160(29.46)
C		2.875(73.03)

Pin Number	Element	Pin Number	Element
1	Spade-0	14	Spade-2
2	Target-9	15	Target-1
3	Target-8	16	Grid Even
4	Grid Odd	17	Target-0
5	Target-7	19	Spade-9
6	Spade-7	20	Spade-8
7	Target-6	21	Heater
8	Target-5	22	Spade-6
9	Spade-5	23	Spade-4
10	Target-4	24	Spade-3
11	Shield Grid	25	Heater
12	Target-3	26	Spade-1
13	Target-2	27	Cathode



PINCIRCLE DIAMETER
OUTER .718 IN.TYPICAL
INNER .437 IN.TYPICAL
PIN DIAMETER .040 IN.
TYPICAL.

FIGURE 1. Outline drawing of electron tube type 6710.

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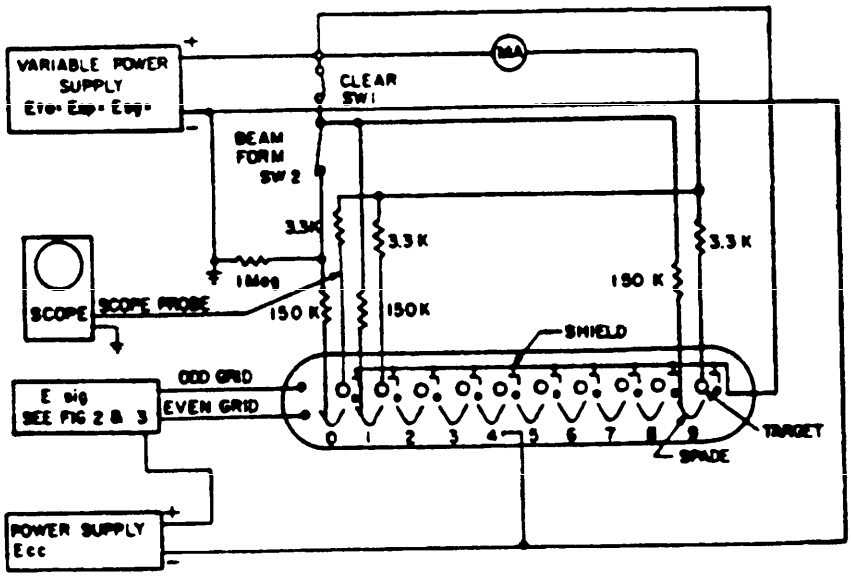


FIGURE 2. Test circuit.

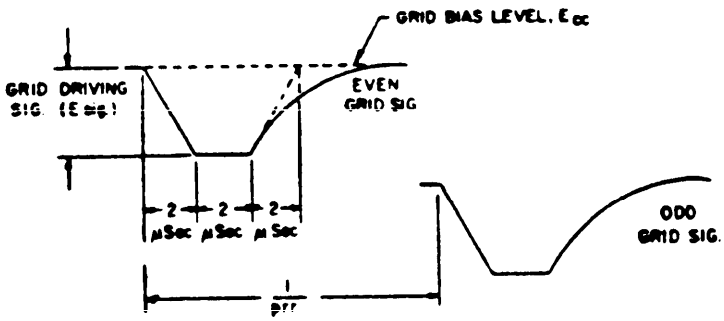


FIGURE 3. Grid signal for counting voltage range (1) and (2).

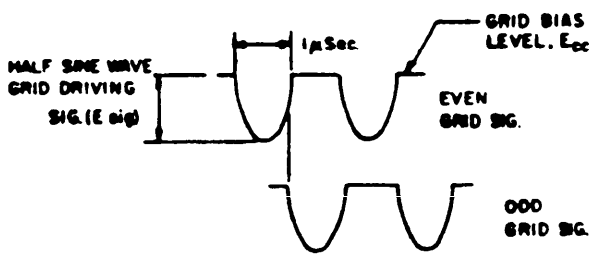


FIGURE 4. Grid signal for counting voltage range (3).

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Rsp - Resistance in series with spade.
Esp - Spade voltage.
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